

TEAM BRIEFING

Rai Mai Phao Zero-Burn Farm

Farmers burn crop residue because it is the **cheapest option**. We make residue **worth money**: our 3-in-1 machine **KhumKha** (Thai for "worth it") in a **village station** turns it into animal feed, fuel briquettes or compost material. Farmers sell through **LINE**, and a **local driver** picks it up at the farm, like Grab.

26.0 M rai burned in Thailand, 2026 (record)	**~1 ฿** per kg we pay for dry residue	**KhumKha** "worth it": our 3-in-1 machine, 85 × 180 cm	**~200k ฿** per machine · rough payback ~17 months

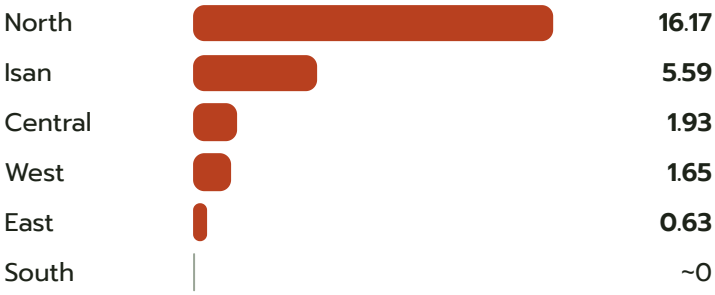
SDG 3 Good health · SDG 8 Decent work · SDG 12 Responsible production · SDG 13 Climate action

> All prices, yields, costs and PM2.5 figures are prototype estimates, to be tested on a real machine. Say so whenever you quote them.

01 · THE PROBLEM

Burning is the cheapest option, so farmers burn

Agricultural area burned, 2026 (million rai)



Source: GISTDA 2026 (via Krungthep Turakij); Rocket Media Lab 2023–2025



North

Largest burned area and the highest lung-cancer rate per person.

Central & West

Corn burning up 760% (Central). Rice burning up 40× (West), Phetchaburi +3,004%.

East

Lung cancer rising fastest, especially Sa Kaeo, the biggest corn province in the region.

Why farmers burn

It is the fastest, cheapest way to clear the field. Ploughing it in costs money and earns nothing, and nobody buys the residue.

Who pays the cost

Everyone: PM2.5 haze, respiratory illness and lung cancer. Each kg not burned keeps about **8 g of PM2.5** out of the air (estimate).

Key line: "Farmers don't burn because they don't know better. They burn because there's no cheaper choice. We make not burning the cheaper choice."

02 · THE SOLUTION

KhumKha: one machine, three products



KhumKha (คุ้มค่า) means "worth it": it makes not burning worth more than burning. 85 × 180 × 60 cm, about a water dispenser. Material flows top to bottom.

HEIGHT	PART	JOB
140–180 cm	Hopper, safety lid, hand guard, magnet, moisture sensor, touch screen, QR scanner, camera	Receive & check
100–140 cm	Hammer-mill shredder ~3 kW + 2 screens	Shred
85–100 cm	Motorised diverter	Route to A/B/C
30–85 cm	A flat-die pellet press · C straight chute · B heated screw press	Process
0–30 cm	3 output trays with scales	Weigh output
Back	Dust fan + bag, control box, vents	Safety

A · PELLET

Animal feed / fuel pellets

Yield **85%**
Time **~60 min / 100 kg**
Fee **1.20 ฿/kg**

B · BRIQUETTE

Fuel briquettes

Yield **80%**
Time **~75 min + 15 warm-up**
Fee **1.50 ฿/kg**

C · CHOP

Compost · mushrooms · mulch

Yield **~90%**
Time **~15 min / 100 kg**
Fee **0.40 ฿/kg**

How it works: 5 steps

1 **Book in LINE**
Residue type & weight; bring it or book a pickup

2 **Village pickup**
A local driver collects at the farm, like Grab

3 **Weigh & check**
Scale, photo and moisture check at the station

4 **Process**
Mode A, B or C

5 **Farmer chooses**
Sell, consign or take home; receipt in LINE



Village station (~2 rai)

Works like a gas station: truck scale, covered drying yard, machine shed, solar roof, lot storage, shop & office, fire safety, parking. Start with **one machine per village**, add more as supply grows.

> Limits: one mode at a time (queue) · flush when switching fuel → feed · spare blades + technician alert every 200 h

03 · PRICES & BUSINESS MODEL

How the money flows

ITEM	PRICE
Buy: corn residue, rice straw, orchard prunings	**1.00 ฿/kg**
Buy: sugarcane leaves	**0.80 ฿/kg**
Moisture ≤ 15% / 15–20% / > 20%	**full / -10% / rejected**
Processing fee if taken home (A / B / C)	**1.20 / 1.50 / 0.40 ฿/kg**
Pickup allowed	**≥ 300 kg or full load**

Pickup fee (radius 10 km)

50 ฿ + 10 ฿/km + 0.10 ฿/kg

Selling to us: farmer pays **≤ 30% of the sale**, our fund pays the rest.
Driver always earns **90%**; platform keeps **10%**.

1,000 kg dry corn residue, 6 km
Sale 1,000 ฿ · pickup 210 ฿ (under 30%)
Farmer nets 790 ฿
Driver 189 ฿ · platform 21 ฿

300 kg, 6 km
Sale 300 ฿ · pickup 140 ฿ → farmer pays 90 ฿, fund pays 50 ฿
Farmer nets 210 ฿
Driver 126 ฿

Business Model Canvas

Key partners
Community enterprises · local government · university · mushroom farms, dairy co-ops, longan dryers · local drivers

Key activities
Buy, process, sell · run the LINE platform · maintain machines

Key resources
3-in-1 machine · LINE system + data · driver network · brand

Value propositions
Farmers: residue earns money, pickup at the farm · Buyers: steady, traceable supply · Community: clean air

Relationships
LINE OA · transparent receipts · public dashboard

Channels
LINE OA + rich menu · drying yard · village leaders

Customer segments
Farmers selling residue · buyers of 6 products · drivers

Cost structure
Machine ~200k ฿ · electricity (solar) · residue purchases · maintenance · labour · pickup fund

Revenue streams
Product sales (main) · processing fees · 10% of pickups · future: CSR, clean-air budgets, carbon credits

Machine cost

~200,000 ฿ (150k–280k)

Shredder 25–40k · pellet press 25–45k · briquette press 40–80k · frame 15–25k · controls 10–20k · screen/camera 5–10k · sensors 4.5–11k · dust/misc 7–13k · assembly 15–30k

Payback (illustration)

400 kg/day × 25 days = 10 t/month
× ~1.2 ฿/kg margin ≈ 12,000 ฿/month
200,000 ÷ 12,000 ≈ **17 months**

04 · IMPACT

Four SDGs, everyone in the chain gains

3
GOOD HEALTH & WELL-BEING
No burning → less PM2.5 → fewer respiratory illnesses and lung cancers

8
DECENT WORK & GROWTH
Farmers earn from residue; villagers earn as drivers

12
RESPONSIBLE PRODUCTION
Residue becomes feed, fuel and compost: nothing is wasted

13
CLIMATE ACTION
Less carbon from open burning, measured on the dashboard

Also: SDG 7 (biomass briquettes + solar roof) and SDG 15 (fewer wildfires, residue back to the soil).

Farmers
Income from what they used to burn, no hauling, no burning fines, transparent LINE receipts.

Village drivers
Side income from vehicles they already own: always 90% of the full fare.

Buyers
Cheap, steady raw material instead of charcoal, firewood or costly feed.

Community
Cleaner air, fewer respiratory illnesses, local jobs.

Local government
A measurable tool to cut burning, with data from every run.

Environment
Less smoke and carbon; residue returns to the soil as compost.

People
Income for farmers and drivers

Planet
PM2.5 and carbon avoided

Profit
Product margin + fees + 10% platform

Measured on every run
kg not burned · PM2.5 avoided (~8 g/kg) · farmer income · driver income · fund paid. All shown on the public dashboard.

Software (working prototype)
LINE OA + LIFF (sell, book pickup, receipts, PDPA consent) · driver app · kiosk screen · dashboard · 3D model. Running cost 0 ฿/month.



05 · WHAT TO SAY

Scripts

30 seconds · passers-by

"Ever been stuck in the smoke haze in the dry season?" *(wait for the nod)*
This year Thailand burned **26 million rai** of crop residue, a record, because burning is the cheapest option. So we built **KhumKha**, Thai for "worth it": a machine the size of a water dispenser that turns crop waste into **feed, fuel briquettes or compost**, all run from **LINE**. Farmers get paid instead of burning, villagers earn as drivers, everyone breathes cleaner air. That's **SDG 3, 8, 12 and 13**.
"Want to try? Scan this QR, it takes 10 seconds."

4 minutes · judges

PART	SHOW	SAY	TIME
Open	Boards	"We're Rai Mai Phao, the Zero-Burn Farm. We make crop waste worth more than burning it."	10 s
Problem	Board 1 map	"26 million rai burned in 2026, a record. The North burned 16 million and has the highest lung-cancer rate. Burning is simply the cheapest option."	30 s
Solution + demo	Board 2 + **iPad in the judge's hands** (Show workflow, X-ray)	"This is KhumKha, Thai for 'worth it'. It shreds residue, then makes feed, briquettes or compost. It weighs, photographs and checks moisture, then sends a receipt to LINE."	50 s
System	Board 3 station	"One station per village, like a gas station. Farmers book in LINE, a local driver collects like Grab. Farmers pay at most 30% of the sale for pickup; our fund covers the rest."	30 s
Business	iPad → Business Model	"Three revenue streams: product sales, processing fees and 10% of pickup fees. A machine costs about 200,000 baht, with a rough payback of 17 months."	40 s
Sustainability	SDG strip, board 2	"SDG 3 cleaner air, 8 income, 12 nothing wasted, 13 less carbon. People, Planet and Profit together."	20 s
Close	Phone with LINE	"Want to try? Sell some residue in LINE, and the receipt arrives in seconds."	20 s

Always add

"These are prototype estimates; our next step is testing a real machine."

Short on time?

Go straight to the iPad demo → Business Model → SDGs.

Greeter

At the front of the booth: "Ever been stuck in the smoke haze?"

Speaker

Next to the boards: follows the script and points at each board.

Demo

Holds the iPad and phone; lets people rotate the model and try LINE.

06 · QUESTION BANK

Likely questions, short answers

PROBLEM & IDEA

Why do farmers burn?
It's the cheapest, fastest way to clear the field. Ploughing it in costs money and earns nothing.

Why not just ban burning?
Bans exist, but without a cheaper alternative farmers still burn. We give them a reason not to: income.

Who is your customer?
Two sides: farmers selling residue, and buyers of the products (mushroom farms, dairy co-ops, animal farms, factories with boilers).

MACHINE

What does it make?
A: feed or fuel pellets · B: fuel briquettes · C: chopped material for compost, mushrooms or mulch.

How fast is it?
~15 min per 100 kg to chop, ~60 for pellets, ~75 for briquettes.

What if the residue is wet?
Above 20% moisture it's rejected; the station has a drying yard.

What if it breaks?
Spare blades, and the system alerts a technician every 200 working hours.

What's the machine called?
KhumKha, Thai for "worth it". The brand is Rai Mai Phao / Zero-Burn Farm.

Is it safe?
Lid cut-off switch, hand guard, magnet, emergency stop, dust extraction.

Is it built?
It's a detailed 3D design and a working software prototype. A pilot machine is the next step.

BUSINESS

How do you make money?
Product margin, processing fees, and 10% of pickup fees.

How much does it cost?
~200,000 ฿ per machine (150k–280k). The station depends on local land and budget.

When does it pay back?
Roughly 17 months at 400 kg/day and ~1.2 ฿/kg margin. An illustration to be tested.

Why 1 baht per kg?
It beats burning (zero income) and leaves margin after processing.

Who pays the pickup fund?
Product margins: pickup is part of our raw-material cost. It only applies when farmers sell to us.

Why 30%?
So pickup never eats most of a small farmer's income.

Stopping tiny pickups?
Pickup needs ≥ 300 kg or a full load; we suggest sharing trips with nearby farms.

Competitors?
Biomass power plants buy residue, but only near the plant and in bulk. We go to the village and to small farmers.

IMPACT, TECH & LAW

Which SDGs?
3, 8, 12, 13 (also 7 and 15).

How do you measure impact?
Every run logs kg not burned, PM2.5 avoided (~8 g/kg), farmer and driver income, on the dashboard.

Is 8 g/kg real?
It's an estimate; we'll replace it with a referenced research value.

Why LINE?
Farmers already use it every day: no new app to install.

Personal data (PDPA)?
Separate consent, only needed data, masked names in public, and farmers can withdraw and delete.

Can you sell the feed?
First in community farms; selling needs registration under Thai feed law.

HARD ONES

Biggest risk?
The numbers are untested. A pilot machine is our next step.

What would you do with funding?
Build one pilot machine in a Chiang Rai village, test yields and costs, then fix the numbers.

What did you learn?
The real problem isn't awareness, it's economics. Make the right choice the cheaper choice.

07 · BEFORE WE OPEN

Checklist & links

10 minutes before

- [ ] iPad on **/ipad/3d**: model loaded, EN, Full screen
- [ ] Monitor on **/present**, press **F**
- [ ] Phone: one test sale in LINE, receipt received
- [ ] Wi-Fi + hotspot backup
- [ ] iPad and phone charged · power strip
- [ ] QR stand at the corner nearest the walkway
- [ ] Rehearse 30 s and 4 min versions with a timer
- [ ] Practise handing the iPad over and switching to Business Model

Links

Website	**sustainable-final.vercel.app**
iPad · 3D model	**/ipad/3d**
iPad · Business Model	**/ipad/bmc**
Presentation	**/present**
Prices & calculator	**/prices**
Dashboard	**/dashboard**
Kiosk demo	**/kiosk**

> The iPad pages are also in the website's Team menu.

Numbers to say out loud

26.0 M
rai burned, 2026

16.17 M
rai in the North

1 ฿/kg
buy price

≤ 30%
max pickup share

90%
to the driver

KhumKha
machine name = "worth it"

~200k ฿
per machine

~17 mo
payback

One sentence to remember: "We make not burning the cheaper choice: farmers earn, drivers get work, buyers get cheap raw material, and everyone breathes cleaner air."